Biology

Topic: Morphology of flowering plants

The Root

1. When a root is seen in vertical section, certain regions can be identified that are arranged one above the other. Can you identify the correct sequence?
(a) Zone of meristems
(b) Root cap zone
(c) Zone of maturation
(d) Root hair zone
(e) Zone of elongation
1) a, d, b, c, e
2) d, b, c, e, a
3) b, a, e, d, c
4) c, b, d, e, a

2. In majority of dicotyledonous plants, the direct elongation of the ___ A ___ leads to the formation of ____ B _____ Select the correct to option to fill the blanks.

	A	B
1)	Radicle	Root hairs
2)	Plumule	Primary root
3)	Radicle	Primary root
4)	Plumule	Adventitious root

3. In which of the following plants, primary root is short lived and new roots originate from the base of stem?
1) Banyan tree
2) Wheat
3) Mango
4) Rhizophora

4. Which of the following is/are the function(s) of a typical root system?
(a) Absorb water and minerals from the soil.
(b) Provide anchorage to plant parts.
(c) Store reserve food material.
(d) Synthesise plant growth regulators.
1) Only (a) and (b)
2) Only (a), (b), and (c)
3) Only (a), (c), and (d)
4) All (a), (b), (c), and (d)

5. Adventitious roots are adventitious in their
1) Position
2) Function

3) Internal structure
4) Place of origin

6. Read the following statements about region of the root and select the incorrect option.
1) Region of meristematic activity is sub-terminal.
2) The cells proximal to the region of meristematic activity undergo rapid elongation and enlargement.
3) The zone proximal to region of elongation is called the zone of cell differentiation.
4) Root hairs arise endogenous from some cells of region of elongation.

7. Tap roots of carrot, turnip and adventitious roots of sweet potato are modified to
1) Provide protection
2) Get oxygen for respiration
3) Store food
4) Provide support

8. Given below are two statements:
Statement I: In some plants, like grass, Monstera and banyan tree, roots arise from radicle.
Statement II: Epidermal cells of region of elongation in root form very fine and delicate, thread like structures called root hairs.
In the light of the above statements, choose the most appropriate answer from the options given below:
1) Both statements I and II are incorrect.
2) Statement I is correct, but statement II is incorrect.
3) Statement I is incorrect, but statement II is correct.
4) Both statements I and II are correct.

9. Select the correct option with respect to Rhizophora.
a. Grows in mangroves or swampy areas.
b. Pneumatophores are present for gaseous exchange.
c. Many roots come out of the ground and grow vertically upwards.
1) (a) and (b) only
2) (a), (b), and (c)
3) (a) and (c) only
4) (b) and (c) only

The Stem

10. Read the following statements about the stem and identify them as true (T) or false (F).
(A) It is the ascending part of plant axis.
(B) It develops from radicle of the embryo of a germinating seed.
(C) Stems are generally green when older but brown when young.

(D) It may preform function of vegetative propagation.

	A	B	C	D
1)	T	F	F	T
2)	T	T	F	T
3)	T	T	F	F
4)	T	F	F	F

11. Which of the following is not the function of stem?

1) Conduction of water, minerals and photosynthates.
2) Storage of food, support, protection and vegetative propagation.
3) Spreading out branches bearing leaves, flowers and fruits.
4) Absorption of water, minerals and photosynthates.

12. The axillary bud of stems is modified into woody straight and pointed structure in

1) Bougainvillea
2) Citrus
3) Turmeric
4) Both (1) and (2)

13. Axillary buds in stems may get modified into

1) Tendrils
2) Thorn
3) Spine
4) Both (1) and (2)

14. Match the following columns and choose the correct option.

Column I (Plant)		Column II (Underground stem modification for food storage)	
1	Potato	a	Rhizome
2	Ginger	b	Tuber
3	Garlic	c	Corm
4	Yam	d	Bulb

1) 1-a, 2-b, 3-d, 4-c
2) 1-b, 2-a, 3-c, 4-d
3) 1-b, 2-a, 3- d, 4-c
4) 1-a, 2-b, 3-c, 4- d

15. Read the following statements and choose the correct option.

Statement A: Underground stems of potato, ginger, turmeric, zamikand, and Colocasia are modified to store food in them.

Statement B: They also act as organs of perennation to tide over conditions unfavourable for growth.

1) Only statement A is correct.
2) Only statement B is correct.
3) Both statements are correct.

4) Both statements are incorrect.

16. Select the option that is correct for the description of subaerial modification.

A. A slender lateral branch arises from the base of the main axis and, after growing aerially for some time, arches downwards to touch the ground.

B. A lateral branch originates from the basal and underground portion of the main stem, grows horizontally beneath the soil, and comes out obliquely upwards, giving rise to leafy shoot.

	A	B
1)	Eichhornia	Mint
2)	Mint	Jasmine
3)	Jasmine	Pineapple
4)	Pistia	Eichhornia

17. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Phylloclades are flattened/cylindrical stem or branches of unlimited growth like in Cactus.

Reason (R): Stems are modified into phylloclade, to facilitate photosynthesis.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

The Leaf

18. A typical leaf consists of three main parts, which are

1) Leaf base, petiole and stipule
2) Leaf base, stipule and lamina
3) Leaf base, petiole and leaf blade
4) Both (2) and (3)

19. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Petioles are long, thin and flexible.

Reason (R): Petioles allow leaf blades to flutter in wind, thereby cooling the leaf and bringing fresh air to leaf surface.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

20. Select the incorrect statement about leaves.

1) Leaf develops at the node.
2) Leaves originate from shoot apical meristem.
3) Leaves bear a bud in its axil.
4) Leaves are arranged in basipetal order.

21. The presence of sheathing leaf base covering the stem partially or wholly is the characteristic of certain
1) Monocots
2) Fern
3) Mosses
4) Legumes

22. Leaf base is swollen to form (A) in some (B). Choose the correct option.

	A	B
1)	Pulvinus	Legumínous plants
2)	Pulvinus	Crucifers
3)	Phyllode	Leguminous plants
4)	Phylloclade	Plants of arid regions

23. Leaf base may bear two small leaf-like structures called
1) Bracts
2) Axillary bud
3) Stipules
4) Prickle

24. Palmately compound leaves
(a) Possess bud in the axile of leaflets.
(b) Possess a number of leaflets attached laterally on the rachis.
(c) Have leaflets attached at a common point, i.e. at the tip of petiole.
(d) Are present in neem.
1) (b) and (d) only
2) (a), (b), and (d)
3) (c) only
4) (a) and (c) only

25. The arrangement of leaves on stem or its branches is called
1) Phyllotaxy
2) Vernation
3) Venation
4) Heterophylly

26. Match the following columns and choose the correct option.

Column I		Column II	
1	Whorled phyllotaxy	a	A single leaf arises at each node in alternate manner
2	Alternate phyllotaxy	b	A pair of leaves arise at each node and lie opposite to each other
3	Opposite phyllotaxy	c	If more than two leaves arise at a node and form a whorl

1) 1-a, 2-c, 3-b
2) 1-a, 2-b, 3-c
3) 1- c, 2-a, 3-b
4) 1-b, 2-a, 3-c

27. Which of the following plants shows alternate phyllotaxy?
1) Nerium, Alstonia, Calotropis
2) China rose, mustard, sunflower
3) Guava, China rose, Calotropis
4) Guava, Calotropis, sunflower

28. Leaf in silk cotton tree is
1) Simple
2) Pinnately compound
3) Palmately compound
4) Ephemeral

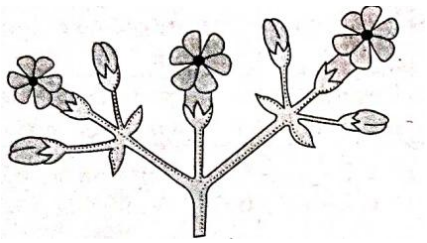
29. Given below are two statements:
Statement I: The leaf is a lateral, generally flattened structure borne on the stem.
Statement II: The shape, margin, apex, surface and extent of incision of lamina varies in different leaves.
In the light of the above statements, choose the most appropriate answer from the options given below:
1) Both statements I and II are incorrect.
2) Statement I is correct, but statement II iş incorrect.
3) Statement I is incorrect, but statement II is correct.
4) Both statements I and II are correct.

Inflorescence and Flower

30. Arrangement of flower on the floral axis is called
1) Inflorescence
2) Placentation
3) Phyllotaxy
4) Vernation

31. In cymose inflorescence
(a) The main axis terminates into flower.
(b) Flowers are born in acropetal manner.
(c) Young flowers are present towards the base and older at the apex.
1) Only (c) is correct.
2) Only (a) and (c) are correct.
3) Only (a) is correct.
4) Only (a) and (b) are correct.

32. Identify the given diagram and choose the incorrect option.



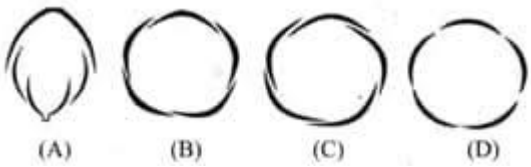
1) The main axis terminates into a flower.
2) Flowers are basipetally arranged.
3) It is a racemose inflorescence.
4) All of the above.

33. When a shoot tip transforms into a flower, it is
1) Always solitary
2) Always racemose
3) Always cymose
4) May be racemose or cymose

34. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Flowers of bean and Cassia have zygomorphic symmetry.
Reason (R): Flowers of bean and Cassia can be divided into two equal radial halves in any radial plane passing through the centre.
In the light of the above statements, choose the most appropriate answer from the options given below:
1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

35. In some flowers such as lily, the calyx and corolla are not distinct and are termed _____
1) Perianth
2) Bract
3) Tepals
4) Peduncle

36. The aestivation in petals of China rose and gulmohar plants can be shown, respectively, by



1) B and A
2) C and B
3) C and D
4) D and C

37. Which type of placentation is found in tomato and Argemone, respectively?

1) Axile and free central
2) Axile and parietal
3) Parietal and axile
4) Marginal and parietal

38. Read the following statements and choose the incorrect one.
1) Gynoecium is the female reproductive part of the flower and is máde up of one or more carpels.
2) The stigma connects the ovary to the style.
3) Placenta is a cushion-like structure within the ovary.
4) Stigma is the receptive surface for pollen grains.

39. Select the correct set of option with respect to position of floral parts on thalamus.

	Flower	**Ovary**	**Example**
1)	Hypogynous	Half inferior	Guava
2)	Epigynous	Superior	Cucumber
3)	Perigynous	Inferior	Ray florets of sunflower
4)	Hypogynous	Superior	China rose

The Fruit and The Seed

40. Consider the following statements and mark true (T) or false (F).
A. The fruits consist of pericarp and seeds.
B. Coconut have fibrous mesocarp.
C. Fruits formed without fertilization are called parthenocarpic.
D. Fruit is mature or ripened ovule.

	A	**B**	**C**	**D**
1)	T	F	F	T
2)	T	T	T	F
3)	T	T	F	T
4)	F	F	T	T

41. Drupe in mango develops from
1) Monocarpellary superior ovary
2) Multicarpellary superior ovary
3) Monocarpellary inferior ovary
4) Multicarpellary inferior ovary

42. Coconut is similar to mango fruit in
1) Being a drupe
2) Absence of endocarp
3) Having fibrous mesocarp
4) None of the above

43. Match the columns.

Column I		Column II	
a	Endosperm	i	Nutrition
b	Aleurone layer	ii	Seed

c	Parthenocarpic fruit	iii	Triple fusion
d	Ovule	iv	Without fertilisation

1) a-ii, b-iv, c-iii, di
2) a-iii, b-i, c-iv, d-ji
3) a-iv, b-iii, c-i, d-ii
4) a-ii, b-i, c-iv, d-iii

44. Mark odd one out with respect to maize grain.
1) Presence of single shield-shaped cotyledon.
2) Aleurone layer separates the embryo from the endosperm.
3) Presence of coleoptile and coleorhiza.
4) Membranous seed coats generally free from pericarp.

45. The large and shield-shaped cotyledon in maize seed is
1) Coleoptile
2) Scutellum
3) Epiblast
4) Coleorhiza

46. Endosperm, a product of double fertilisation in angiosperms is absent in the seeds of
1) Coconut
2) Wheat
3) Maize
4) Orchids

47. Generally monocotyledonous seeds are (A) but some as in (B) are (C).

	A	**B**	**C**
1)	Non-endospermic	Orchids	Endospermic
2)	Endospermic	Orchids	Non-endospermic
3)	Non-endospermic	Castor	Endospermic
4)	Endospermic	Castor	Non-endospermic

48. Scar on the seed coat through which the developing seeds were attached to the fruit is called
1) Testa
2) Tegmen
3) Hilum
4) Raphe

49. The embryo is small and situated in a groove at one end of the endosperm in
1) Monocot
2) Dicot
3) Both monocot and dicot
4) All gymnosperms and angiosperms

Semi-Technical Description of a Typical Flowering Plant and Description of Some Important Families

50. Which of the following floral features is not represented by symbols in the floral formula of a plant family?
1) Symmetry of flower
2) Aestivation of calyx and corolla
3) Cohesion and adhesion between floral whorls
4) Relative position of ovary with respect to other parts

51. The character of flower represented by floral formula but not by floral diagram is
1) Adhesion of stamen
2) Placentation
3) Position of gynoecium
4) Aestivation

52. In the floral formula, $K_{(n)}$ denotes _____ condition
1) Polysepalous
2) Gamosepalous
3) Episepalous
4) Epiphyllous

53. Read the following statements and select the correct option.
Statement A: In the floral formula, a superior ovary is represented by a line drawn above G.
Statement B: The position of the mother axis with respect to the flower is represented by a dot below the floral diagram.
1) Only statement (A) is correct.
2) Only statement (B) is correct.
3) Both statements (A) and (B) are correct.
4) Both statements (A) and (B) are incorrect

54. Which of the given plants, used as a medicinal plant is of family Leguminoceae?
1) Mulethi
2) Petunia
3) Ashwagandha
4) Aloe

55. Which of the given features is not related to mustard?
1) Tetradynamous condition
2) Hypogynous flower
3) Twisted aestivation
4) Parietal placentation

56. Read the following characters and mark the correct ones for family Solanaceae.
a. Syncarpous superior ovary.
b. Actinomorphic flowers.
c. Swollen placenta.
d. Epiphyllous condition.
e. Endospermic seeds.
1) Only (a) and (e)

2) Only (a), (b), (c), and (e)
3) Only (a), (c), and (d)
4) Only (c), (d), and (e)

57. Persistent calyx is the character of plants belonging to the family
1) Malvaceae
2) Cruciferae
3) Leguminoceae
4) Solanaceae

58. Diadelphous condition, vexillary aestivation, and marginal placentation are diagnostic features of plants belonging to
1) Cotton family
2) Pea family
3) Potato family
4) Mustard family

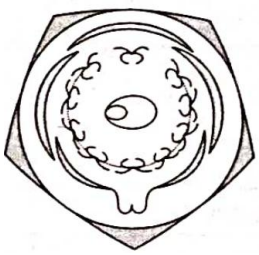
59. How many of the following plants belong to the family whose formula is given below?

$\%\,⚥\,K_5C_{1+2+(2)}\,A_{(9)+1}\underline{G}_1$

Soybean, groundnut, Indigofera, chilli, Solanut, Trifolium, Sesbania, tulip, sun hemp, Colchicum, Asparagus, Petunia

1) Six
2) Five
3) Seven
4) Four

60. Select the characteristic feature of plants that has the given floral diagram.



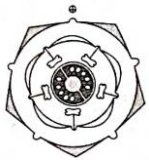
1) Bisexual actinomorphic flower
2) Monocarpellary unilocular ovary with many ovules
3) Epigynous flower with marginal placentation
4) Monadelphous condition of androecium

61. Floral diagram of the family to which cereals and millets belong is

1)

2)

3)

4) 

62. Select the incorrectly matched pair.

1)	Sunflower	$\widehat{K_{\text{o or p}}\ C_{(5)}}$
2)	Brassica	K_{2+2}
3)	Soyabean	$A_{(9)+1}$
4)	Rice	A_2

63. Read the following statements and choose the incorrect one.

1) To describe a flowering plant, various morphological features are used.
2) The plant is described beginning with its habit, vegetative characters - roots, stem and leaves and then floral characters inflorescence and flowering parts
3) A floral diagram and a floral formula are presented before describing various parts of a plant.
4) All statements are correct

64. Which of the following floral formula symbol denotes actinomorphic and zygomorphic symmetry?

1) % and ⊕ respectively
2) ⊕ and % respectively
3) A and Z respectively
4) ♀ and ♂ respectively

65. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): In floral formula, (K) denotes gamosepalous condition.

Reason (R): In floral formula, fusion is indicated by enclosing the figure within bracket.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

66. Match the following column and choose the correct option.

Column I	Column II

1	Cruciferae	a	% ⚥ $K_{(5)}C_{1+2+(2)}A_{(9)+1}\underline{G}_{(1)}$
2	Leguminoceae	b	⊕ ⚥ $K_{2+2}C_4A_{2+4}\underline{G}_{(2)}$
3	Gramineae	c	Br.% $K_{pappus}C_{(5)}A_0\bar{G}_{(2)}$
4	Compositae	d	Br.% ⚥ $P_2A_3\underline{G}_1$

1) 1-b, 2-a, 3-c, 4-d
2) 1-a, 2-b, 3- c, 4-d
3) 1-b, 2-a, 3-d, 4-c
4) 1-a, 2-b, 3-d, 4-c

67. The family Cruciferae is characterised by
1) Bicarpellary, syncarpous, superior, unilocular ovary with parietal placentation
2) Bicarpellary, apocarpous, superior, trilocular ovary with axile placentation
3) Tricarpellary, syncarpous, inferior, bilocular ovary with marginal placentation
4) Tricarpellary, apocarpous, inferior, bilocular ovary with axile placentation

68. Hibiscus, cabbage and marigold plants belongs to family
1) Malvaceae, gramineae and compositae respectively
2) Malvaceae, Brassicaceae and Assteraceae respectively
3) Compositae, malvaceae and gramineae respectively
4) Gramineae, solanaceae and leguminoceae respectively

69. Blue dye is extracted from _____
1) Indigofera
2) Cassia
3) Cinchona
4) Eucalyptus

70. The floral formula, (C) denotes
1) Polysepalous
2) Polypetalous
3) Gamopetalous
4) Gamosepalous

71. In Leguminoceae family, which of the following is correct about gynoecium?
1) Syncarpous; ovary superior, bilocular, placenta swollen with many ovules, axile placentation
2) Ovary superior, monocarpellary, unilocular with many ovules, style single
3) Ovary inferior, bicarpellary, unilocular with many ovules, style single
4) Apocarpous; ovary superior, unilocular, placenta swollen with many ovules, axile placentation

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
3	3	2	4	4	4	3	1	2	1	4	4	4	3	3
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
3	4	3	4	4	1	1	3	3	1	3	2	3	4	1
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
2	3	1	2	1	2	2	2	4	2	1	1	2	4	2
46	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
4	2	3	1	2	3	2	4	1	3	2	4	2	1	2
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**	**71**				
1	4	3	2	4	3	1	2	1	3	2				